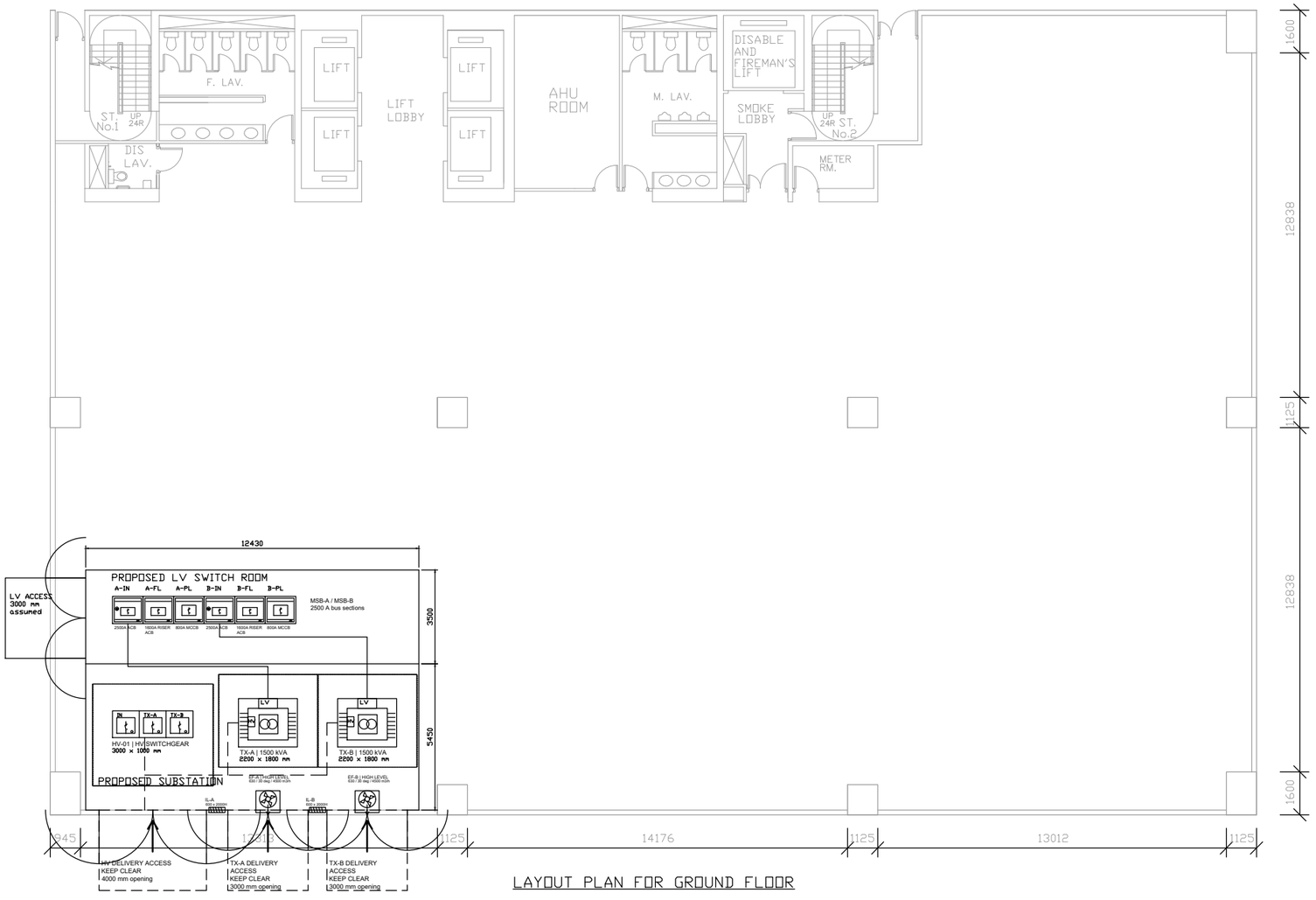
TASK 1 | GROUND-FLOOR PLANTROOM LOCATION

A3 LANDSCAPE | SCALE 1:200 | DIMENSIONS IN mm | GROUND FLOOR



PROPOSED LOCATION
Lower-left perimeter of supplied ground-floor plan.
Substation 12430 x 5450; adjacent LV room 12430 x 3500 assumed. See separate 1:75 room detail.

LOCATION REASONS
Direct external plant access is proposed, subject to site confirmation.
Adjacent LV room keeps transformer-to-incomer interfaces short.
Perimeter plantrooms separate supply equipment from the upper lift/stair/AHU core and remaining accommodation.
LV room has its own proposed west entrance.

LOAD ASSESSMENT
Offices: 1644.800 kVA.
AC: 1000 kW cooling -> 526.039 kW electrical -> 584.488 kVA at assumed PF 0.90.
Public services: 200 kVA aggregate allowance.
Two-zone conservative envelope: 2672.216 kVA.
Propose 2 x 1500 kVA, with equal zone allocation; no full-load N-1.

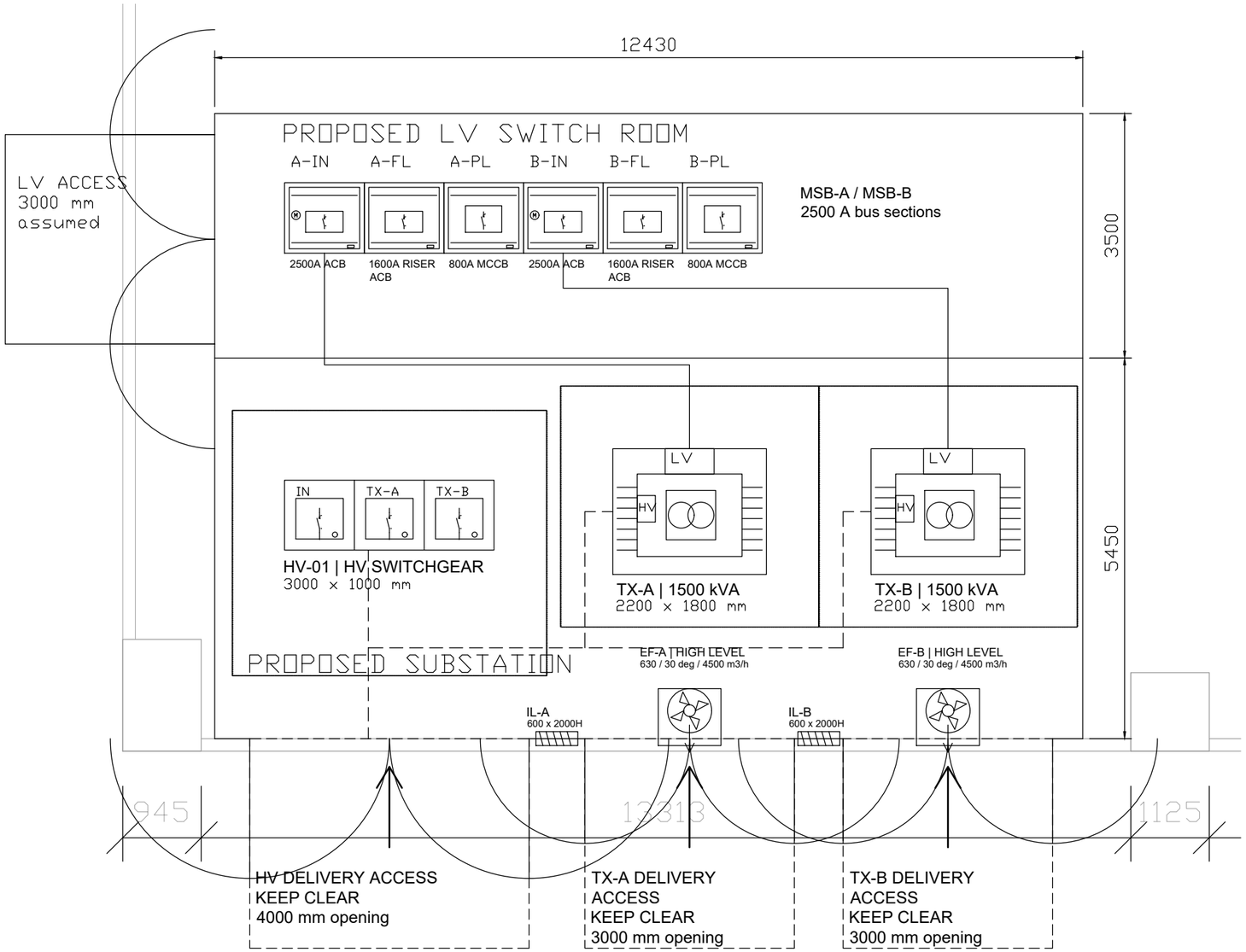
SITE CONDITIONS TO VERIFY
External apron, road and <=20m unloading route are not given by the source plan. Maintain >=3m clear delivery route.
Architectural wall openings, structural/flood/fire provisions, ventilation and supplier equipment approval require coordination.

GROUP 1 | TASK 1

Design basis: CLP COP215 Rev07; CLP COP101 v16.

TASK 1 | PROPOSED SUBSTATION AND MAIN LV ROOM

A3 LANDSCAPE | SCALE 1:75 | DIMENSIONS IN mm | PLANTROOM DETAIL



LINE KEY: dashed path = HV trench; solid path = LV route. Dashed boxes = delivery clearance; arrows = delivery direction. Equipment details diagrammatic.

LV INTERFACES / HV ROUTES

LV trench: 1000 W x 1100D, four MCTs per interface. A: 8.03+2=10.03m; B: 8.31+2=10.31m (<20m). 2m is assumed total termination/vertical allowance.

HV centre-lines: 1000 W at panels; 600 W branches; final 300 W; generally 1400D. Segregate HV/LV. Covered trenches; no beam obstruction.

DETAIL 1:75 AT A3

CAPACITY AND EQUIPMENT

TX-A/B: CLP 1500 kVA distribution transformer type; 3000 kVA installed, 2672.216 kVA demand. No full-load N-1.

HV-01: incomer + two transformer protection panels.

MSB-A/B: 2500 A sections. IN: MTZ2 25H1; FL: MTZ2 16H1 rising-main ACB; PL: NS800H. Floor taps: NSX400H MCCBs.

Plan outlines are installation design dimensions; utility equipment supply follows COP101 responsibilities.

CLEARANCES / BUILDING WORKS

HV rear 1000, front 1800, sides 750. TX LV terminals north: 900; other sides 750.

Proposed 2h FRR partitions; fire-seal penetrations. FFL +0.15 vs external +0.00 assumed.

Actual structures, wall thickness and equipment dimensions require coordination.

ACCESS AND DOORS

South external access assumed. Provide 3000-wide clear delivery route; unloading distance <=20 m to verify.

Plant doors: HV 4000 W; TX 3000 W each; 2800 H assumed. Outward opening, shown at 90 degrees.

Adjacent plant doors used sequentially; TX-A has wicket personnel access.

Independent LV entrance west.

VENTILATION

EF-A/B: COP101 v16 6.6 type 630 mm / 30 deg / 700 rpm; each 4500 m3/h at 70 Pa, <=58 dB(A) at 1 m.

Design heat: 2 x 15 kW x 1.25 = 37.5 kW. Combined 9000 m3/h gives 12.44 K air rise.

IL-A/B: each 600 W x 2000H, minimum 50% free area; combined 1.20 m2 >=1.12 m2.

High-level ducts: 900 x 750 mm; discharge >=2.5 m above street, away from people. See ventilation calculation.

CLP COP215 Rev07; COP101 v16, 4.3 and 6.1-6.7.

Equipment selections and calculations accompany this drawing.

TASK 1 | MAIN ELECTRICAL DISTRIBUTION SCHEMATIC

A3 LANDSCAPE | NOT TO SCALE | 2 x 1500 kVA | 380/220 V LV DESIGN BASIS

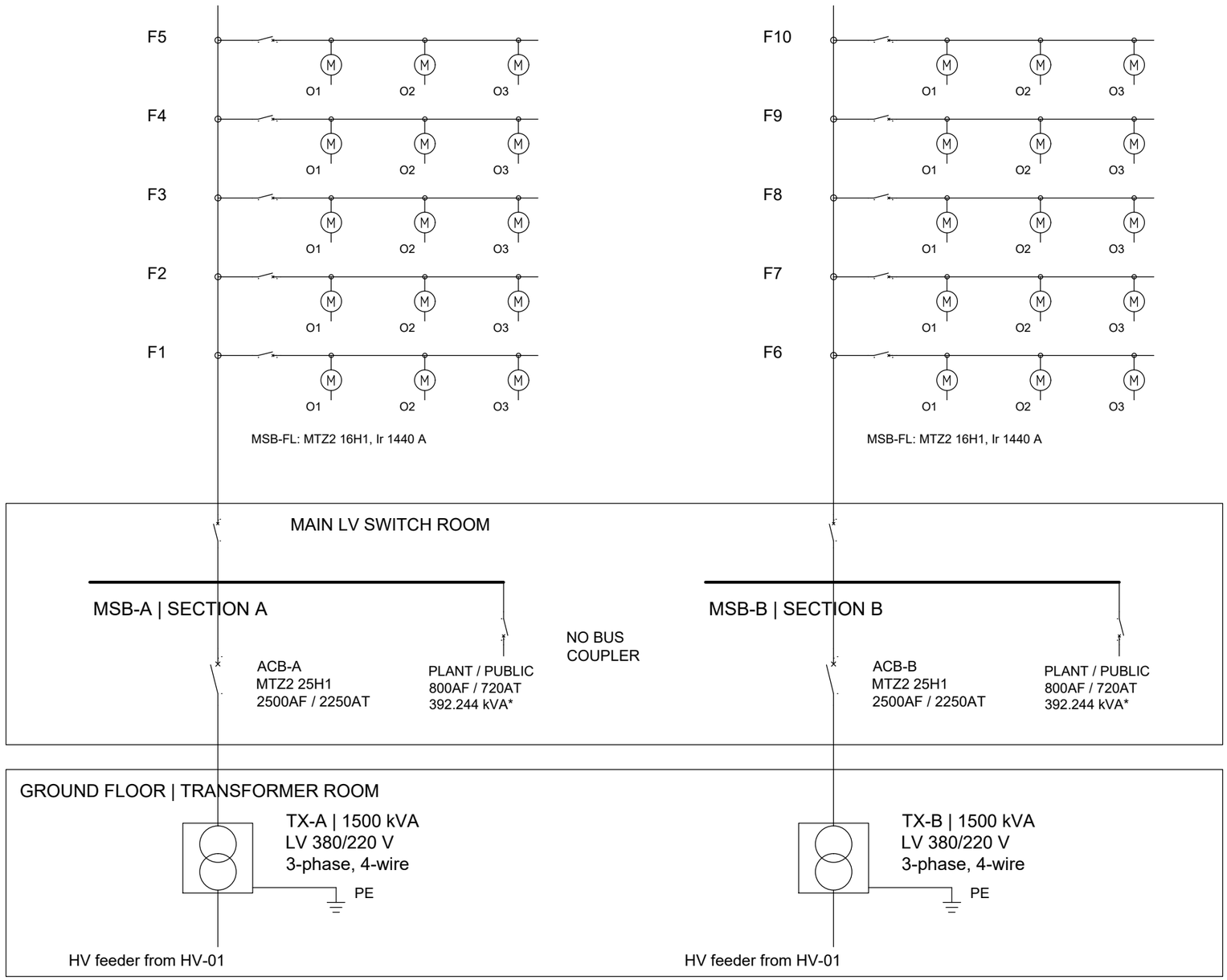
NORMAL SUPPLY: 2672.216 kVA total; two independent LV sections.
Equal allocation: 1336.108 kVA / approx. 2030 A per section.

ZONE A | FLOORS 1-5

1600 A rising main; floor MCCBs 400AF / 320AT

ZONE B | FLOORS 6-10

1600 A rising main; floor MCCBs 400AF / 320AT



* Per section: AC 292.244 + public services 100 = 392.244 kVA before the 1.1 zoning factor.

SYSTEM BASIS

MSB design PFC 51.67 kA: TX uk=6%, -10% tolerance, c=1.1, plus 8 x plant/public current. Independent sections. Main/riser ACBs 66 kA; floor/plant MCCBs 70 kA. AF=frame; AT=long-time setting. Utility supply and protection follow COP101.

REFERENCE: Electrical Schematic 1 - transformer, main switchboard and floor-metering arrangement.
Read with the plantroom layout and Office 1 distribution-board schematic.

TYPICAL OFFICE 1 SUPPLY

Selected tenant detail; applicable to its floor supply zone.

OFFICE 1 | FROM ITS FLOOR METER

M

Office 1 tenant meter

MS-O1 | C63, 4P
A9F54463 | 10 kA

4C35 mm2 PVC Cu
+ CPC16 mm2
31.512 m design length

DB-O1 | 100 A, 4P isolator
15 outgoing RCBO circuits
7 lighting + 8 small power
Ib(max phase) = 48.600 A

P7: WH01 via FS01
SHC10 GB | 1.4 kW
B10 / 10 A DP FCU

TN-S; dedicated N and PE. PE bypasses switching.
Design source: Ze = 0.20 ohm; PFC <=6 kA at MS-O1; upstream drop allowance <=2.2 V.

DESIGN LIMITS

No full-load N-1. Central plant/public load is split equally as a design assumption. This sheet shows normal supply; emergency systems are separate.

GROUP 1 | E-04 | REV 02 | 10 NOV 2026